



Tectonic setting and Pan-African structural evolution of the western Saldania Belt, South Africa

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Abstract

The western Saldania Belt records the closure of the Adamastor Ocean and amalgamation of southwest Gondwana in the latest Neoproterozoic and early Phanerozoic, but the geodynamic setting and evolution of the belt remains controversial. Regional mapping and the integration of structural data presented in this paper document the juxtaposition of domains with distinct fabric development and kinematics. Structures record Pan-African strains and a strike-slip dominated sinistral transpression during southeast-directed subduction and convergence at low angles to the continental margin. Two main, structurally overlying domains – an upper and a lower domain – can be distinguished. Rocks of the upper domain record the partitioning of the transpressional strain into (1) broad regions of more or less upright, north to northwesterly-trending folds (F_2) that record horizontal, mainly coaxial, east-northeast to west-southwest directed shortening at high angles to the continental margin, and (2) the northwesterly-trending, sinistral strike-slip Colenso Fault Zone that accommodates the margin-parallel, non-coaxial component of deformation between *ca.* 555 and 520 Ma. The strike-slip partitioning was promoted by strain localisation into synkinematic plutons of the Cape Granite Suite. In contrast, rocks in the pervasively transposed, *mélange*-like lower domain in the central Swartland region record episodes of vertical, coaxial shortening alternating with non-coaxial top-to-the-west and -northwest thrusting. The coaxial shortening strains relate to the thickening and gravitational loading of the succession in response to thrusting and tectonic underplating. Non-coaxial strains denote episodes of seismic slip during underthrusting.

The different fabric domains are interpreted to represent a section through a fore-arc region, from the deeper level accretionary prism (lower domain) to the overlying, folded fore-arc basin succession (upper domain). Regional-scale klippen structures indicate the transfer of lower-domain phyllites into the overlying fore-arc sequence. The proposed fore-arc setting of the Saldania Belt suggests the subduction of the Adamastor Ocean below the leading edge of the Kalahari Craton. This challenges stratigraphic and structural correlations between the Saldania Belt and the Gariep Belt to the immediate north.

Introduction

The western branch of the Saldania Belt in South Africa forms the southernmost extension of the system of Pan-African and Brasiliano belts in southern Africa and South America that accommodated the amalgamation of southwest Gondwana in the latest Neoproterozoic and early Phanerozoic (Almeida et al., 1973; Porada, 1979, 1989; Torquato and Cordani, 1981; Hartnady et al., 1985; Basei et al., 2000, 2018; Caby 2003; Cordani et al., 2013; D'Agrela-Filho and Cordani, 2017; Konopásek et al., 2020). On the African side, the chronology and kinematic evolution of

Pan-African events are well documented in the northern Kaoko and Gariep belts (Hartnady et al., 1985; Gresse, 1994; Gresse et al., 2006; Frimmel, 1995, 2000, 2018; Frimmel et al., 1996, 2002, 2011; Goscombe et al., 2003, 2005a, 2005b; Goscombe and Gray, 2007, 2008; Frimmel and Fölling, 2004; Konopásek et al., 2005, 2014). The belts record the southeast-directed accretion of terranes onto the cratonic margins along major thrusts and/or steep transpressional shear zones (Von Veh, 1993; Gresse, 1994; Hålbich and Alchin, 1995; Goscombe et al., 2005a; Konopásek

et al., 2005; Goscombe and Gray, 2007). Sinistral transpressional kinematics suggest the oblique convergence and collision of terranes, while magmatic and metamorphic ages chronicle the migration of tectonics from north to south between *ca.* 590 and 545 Ma (Da Silva et al., 2000; Goscombe et al., 2005a; Chemale et al., 2011; Rapela et al., 2011; Kisters et al., 2015).

The tectonic evolution of the Saldania Belt, in contrast, is only poorly understood. For the most part, the belt is developed as a low-grade metamorphic, northwesterly-trending fold belt. The rocks are collectively referred to as the Malmesbury Group, a mainly turbiditic, late-Ediacaran (570 to 550 Ma) sequence with locally intercalated mafic to intermediate volcanic rocks (Gresse et al., 2006). Voluminous granites of the Cape Granite Suite (CGS) intruded shortly after deposition of the Malmesbury Group between *ca.* 550 and 510 Ma (Rozendaal et al., 1999; Gresse et al., 2006; Scheepers and Schoch, 2006; Villaros et al., 2009; Frimmel et al., 2013). However, the very poor outcrop conditions, relatively uniform, low grades of metamorphism, and the deceptive monotony of the rocks have led to often contradicting correlations of units and conflicting interpretations of the internal structure of the belt. This lack of a regional-scale and generally accepted framework for the belt leaves correlations with adjoining belts tentative at best. As a result, many tectonic models for the amalgamation of southwest Gondwana do not even consider the Saldania Belt in their regional reconstructions or merely regard the belt as an extension of the Gariep belt to the north (e.g., Goscombe and Gray, 2008; Gaucher et al., 2009; Miller et al., 2009; Frimmel et al., 2011, 2013; Rapela et al., 2011).

To date, structural studies in the Saldania Belt have mainly focused on smaller and reasonably well-exposed areas, mainly coastal stretches or selected inland regions (e.g., Von Veh, 1983; Belcher, 2003; Rowe et al., 2010). Smaller exposures throughout the region have only been documented in older maps (e.g., Rabie, 1948, 1974; De Villiers et al., 1954; Hartnady et al., 1974; Theron et al., 1992) and the integration of these areas into a coherent geodynamic model and discussion before the background of more recent geochronological data (e.g., Armstrong et al., 1998; Scheepers and Schoch, 2006; Villaros et al., 2009; Frimmel et al., 2013; Kisters et al., 2015) remains tentative. To this end, the present study integrates the structural analysis of outcrops in the belt with historical structural data over an area of some 7000 km², from Cape Town and Strand in the south to Piketberg and Vredenburg in the north. The main aim of the study is to provide a regional compilation of structures and structural domains that can be used to formulate a kinematic framework for the western Saldania Belt. This framework can then be used to place the Saldania Belt into the larger geological context of southwest Gondwana.

Regional geology

Traditionally the Saldania Belt has been interpreted to be underlain by one or two distinct terranes, or tectonostratigraphic domains, thought to have been accreted to the Kalahari Craton along prominent strike-slip fault zones (Hartnady et al., 1974; Rozendaal et al., 1999; Frimmel, 2009; Frimmel et al., 2011, 2013;

Gresse et al., 2006; Buggisch et al., 2010). Most of these models also suggest a close correlation of lithologies and structures with belts to the north and terrane accretion and sedimentation of the Malmesbury Group are essentially regarded as being contiguous with the Gariep Belt as an inverted back-arc basin (Basei et al., 2005; Frimmel et al., 2011, 2013). Field-based studies, in contrast, question the presence of distinct terranes and highlight:

- similar lithological assemblages and age structures of the rocks,
- the uniformly low grades of metamorphism,
- lack of major structural breaks across purported terrane-bounding strike-slip faults, and
- that syn- to late-tectonic CGS plutons have no equivalent in belts to the north (Figure 1) (Rowe et al., 2010; Kisters and Belcher, 2018).

In these models, sedimentation, tectonism and plutonism are related to the deformation of a fore-arc sequence along a convergent margin situated along the leading edge of the Kalahari Craton (Belcher and Kisters, 2003; Rowe et al., 2010; Kisters and Belcher, 2018). Detrital zircon ages from throughout the belt indicate similar, late Neoproterozoic (Ediacaran, *ca.* 560 Ma, Armstrong et al., 1998; Frimmel et al., 2013) ages of sedimentation throughout much of the belt. These ages compare well with U-Pb zircon crystallisation ages of 555 ± 5 Ma for intercalated tuff units (Kisters et al., 2015). The ages also highlight that granites of the CGS have intruded within less than 5 Ma after the deposition, folding, and regional metamorphism of the Malmesbury Group.

This paper builds on the findings of earlier regional studies that have delineated three main structural domains in the western Saldania Belt (Newton, 1966; Hartnady et al., 1974; Belcher, 2003; Belcher and Kisters, 2018). This includes:

- a structurally lower domain dominated by pervasively foliated quartz-sericite phyllites and structurally imbricated metamafic chlorite schist, limestone, graphitic limestone and chert in the central parts of the belt;
- a structurally higher and regionally widespread, folded and well-bedded, largely metaturbiditic greywacke-shale sequence, and
- two northwest-southeast trending fault zones that record mainly strike-slip, but also late-stage normal kinematics, including the more prominent Colenso Fault Zone (CFZ) in the southwest, and the Piketberg-Wellington Fault in the northeast (Figure 1).

Syn-, late- and post-kinematic S-, I- and A-type granites of the Cape Granite Suite (CGS) have intruded the metasedimentary sequence between *ca.* 550 and 510 Ma and underlie probably 30% of the surface area of the Saldania Belt (e.g., Scheepers, 1995; Stevens et al., 2007; Villaros et al., 2009; Farina et al., 2012).

Structural domains of the western Saldania Belt

Lower Phyllitic Domain

Lithological inventory

Quartz-sericite phyllites form the structurally deepest exposed levels of the Saldania Belt (Newton, 1966; Hartnady et al., 1974;

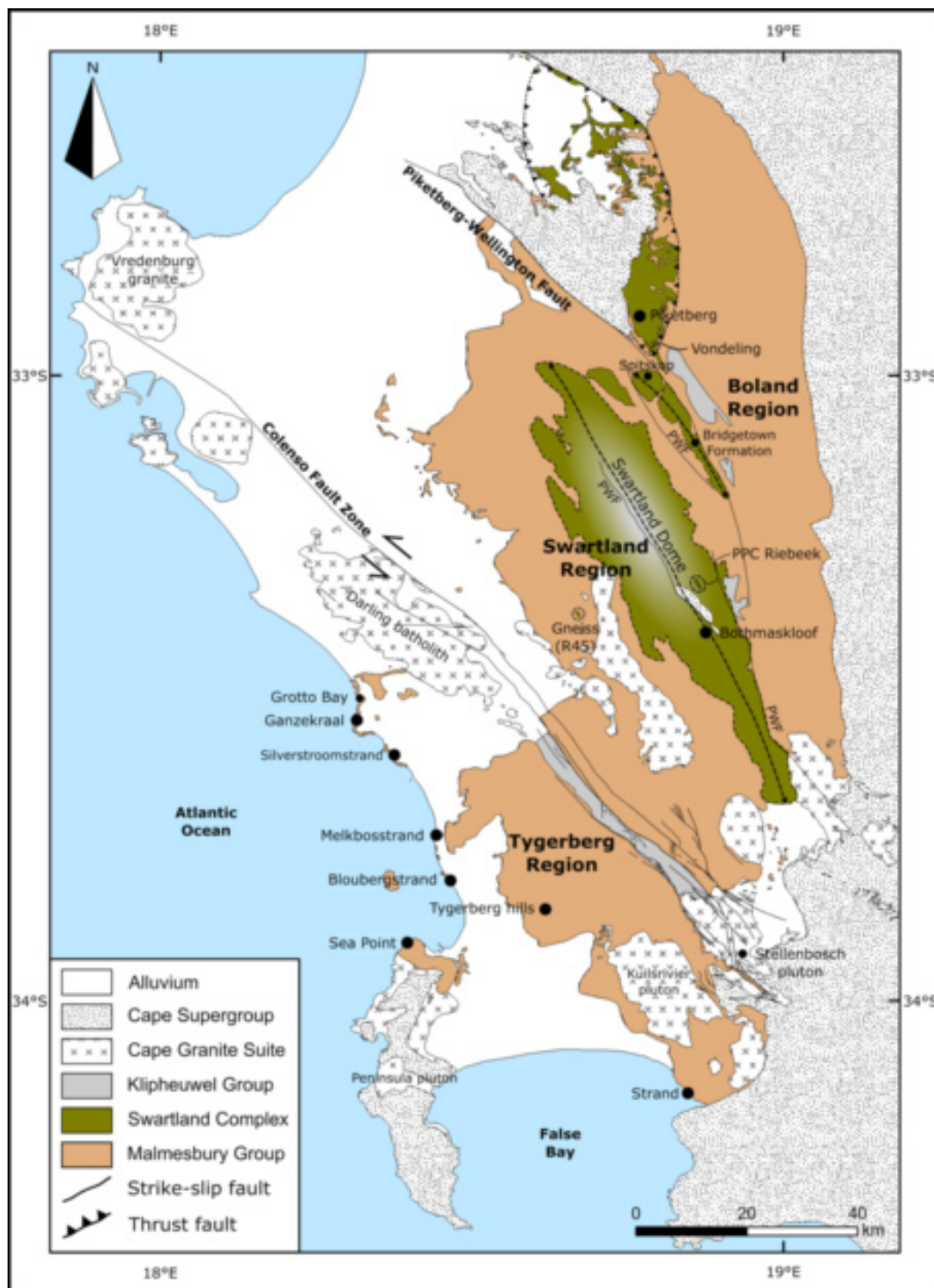


Figure 1. Simplified geological map of the western Saldania Belt showing the distribution of rocks of the upper and lower domains (Malmesbury Group and Swartland Complex) as discussed in this paper. Localities mentioned in the text are also indicated. PWF=Picketberg-Wellington Fault; PPC=Portland Pozzolana Cement quarry operations.

Belcher, 2003; Belcher and Kisters, 2003). Previous works have subdivided the sequence into distinct formations, based on subtle lithological differences (quartz-sericite schist, quartz-schist, quartz-sericite-chlorite schist, quartz-feldspar schist, etc.; e.g., Rabie, 1948; Hartnady et al., 1974; SACS, 1980; Gresse et al., 2006), but this only allows for a very uncertain correlation of units on a regional scale. Bedding in the phyllites is only locally preserved and has been largely overprinted by a bedding-parallel foliation (S_1) (see below). Grey limestones and associated graphitic limestones can intermittently be traced for several tens of kilometres in the phyllites on the eastern limb of the first-order (F_2) Swartland dome (Figure 1). The carbonate rocks are mainly in structural contact with the phyllitic sequence, mined in a number of quarry operations where they form up to 100 m-thick lens-like units or semi-continuous horizons bound by phyllites. Altered metamafic rocks occur as greenish chlorite schists that are, in places, associated with strongly foliated and lineated cherts. Chlorite schist units are common, but volumetrically subordinate since they mainly occur as discontinuous lenses intercalated with quartz-sericite phyllites (Rozendaal et al., 1999). By far the largest metamafic sliver has been distinguished as the Bridgetown Formation (Figure 1) (Hartnady et al., 1974; Gresse et al., 2006). The Bridgetown Formation comprises pervasively altered chlorite-tremolite-carbonate schists and serpentinite ('greenstones') with a tholeiitic geochemical signature (Slabber, 1995; Gresse et al., 2006). The greenstone lithologies are foliated (S_1) and structurally imbricated with chert, graphitic schist, dolomite, and slivers of quartz-sericite phyllite. Large exposures of biotite gneiss, some 8 km northwest of the town of Malmesbury, document the presence of higher-grade metamorphic rocks in the region (Figure 1). Contacts with the lower-grade phyllites are not exposed, but the pervasive fabric development and tight intrafolial folds indicate the affinity of the gneisses to the phyllitic domain, probably representing a deeper, thrust sliver (e.g., Belcher and Kisters, 2003; Frimmel et al., 2013). Given the degree of structural imbrication and the mélangé-like character of the sequence, Kisters and Belcher (2018) have referred to rocks of the phyllitic domain as the Swartland Complex, emphasizing the primarily tectonic contacts between units (Figure 2A).

The basement to the phyllitic sequence is not exposed. However, metasedimentary xenoliths in the Darling batholith contain high-grade metamorphic assemblages (P up to 8 kbar, T ca. 735°C; Villaros et al., 2009) that indicate metamorphism of the rocks at depths of >20 km. The youngest detrital zircon ages of ca. 570 Ma document a Pan-African age of deposition and suggest the depth extent and thickness of the sedimentary protoliths underlying the Saldania Belt (Kisters and Belcher, 2018).

Structural geology

The most characteristic fabric element of the lower phyllite domain is a pervasive phyllitic cleavage (S_1). The cleavage is defined by the preferred orientation of sericite/muscovite and imparts an almost diagnostic silver sheen on the rocks. Millimetre- to metre-scale isoclinal, intrafolial folds testify to the transposition of bedding into the S_1 cleavage (Figure 2B). Bedding (S_0) is only rarely preserved and marked by the

presence of quartzitic units or rather gradual variations of quartz, feldspar, or white mica. On a regional scale, the S_1 cleavage has been refolded and steepened by northerly-trending F_2 folds (see below). F_2 folds dominate the geology of the Saldania Belt and cause the refolded S_1 foliation patterns in, e.g., the central F_2 Swartland dome (as seen on the stereonet, all stereonet are plotted in the lower hemisphere with equal area projections) (Figure 3). In addition to the S_1 phyllitic cleavage, linear fabrics are largely confined to lower domain rocks and almost absent from the upper domain (Figure 3). Most lineations are intersection lineations ($S_0/S_1/S_2$) or crenulations (S_1 crenulated by F_2 folds), which account for the unidirectional trends of lineations parallel to F_2 folds (Figure 3). Shallow plunging northwest-southeast directed L_1 stretching lineations are developed in D1 mylonitic shear zones (see below), but also throughout the region on S_1 foliation planes (Figure 3).

The structural complexity of the lower domain is best appreciated in limestone quarries that provide three-dimensional exposures. In contrast to the pervasive foliation development in phyllites, strain in limestones is more localised. Bedding is well-preserved and deformation is confined to metre-wide, closely spaced (2 to 10 m) bedding-parallel calc-mylonites. Metre-thick, lensoid phyllitic units are structurally interleaved with limestones along mylonitic shear zones, underlining the structural nature of many of the contacts (e.g., Figure 2A). Strain is more distributed in pervasively foliated graphitic limestones that are structurally imbricated with the main, grey limestones. The boudinage of massive limestone units in graphitic limestones indicates a component of vertical, bedding-normal shortening (Figure 2C). Metre-scale duplex structures in graphitic limestones, S-C fabrics (Figure 2D), asymmetric folds, and quartz-carbonate aggregates forming sigma clasts in calc-mylonites (Figure 2E) consistently indicate a top-to-the-west and -northwest sense of shear, parallel to the L_1 stretching lineation. Narrow, brittle, east-northeast-trending normal faults, cross-cut the earlier fabrics (S_0 , S_1 , and mylonitic foliation) at high angles with dips of 40 to 60° to the north and south. Fault planes are lined by quartz or quartz-carbonate veins and the mainly dip-slip displacements are between 2 to 6 m. Despite the cross-cutting nature of the faults, quartz veining and associated chlorite alteration suggest a relatively early timing during the main Pan-African tectonism, and the normal movement points to a component of layer-parallel extension (Figure 3).

Hydrothermal veining

Quartz-, quartz-carbonate-, or carbonate veins may locally constitute up to 10 vol.% of individual outcrops and testify to the voluminous and syntectonic fluid flow through the phyllitic sequence. Vein systems are useful for the analysis of regional stresses as the veins record infinitesimal strain increments so that vein geometries and orientations can be correlated with the stress fields responsible for vein formation (e.g., Robin and Cruden, 1994; Fossen and Tikoff, 1998; Fossen, 2010). In phyllitic units, quartz veins are almost invariably transposed (isoclinally folded and/or stretched and boudinaged) into the S_1 foliation. Limestones contain planar and seemingly undeformed veins,

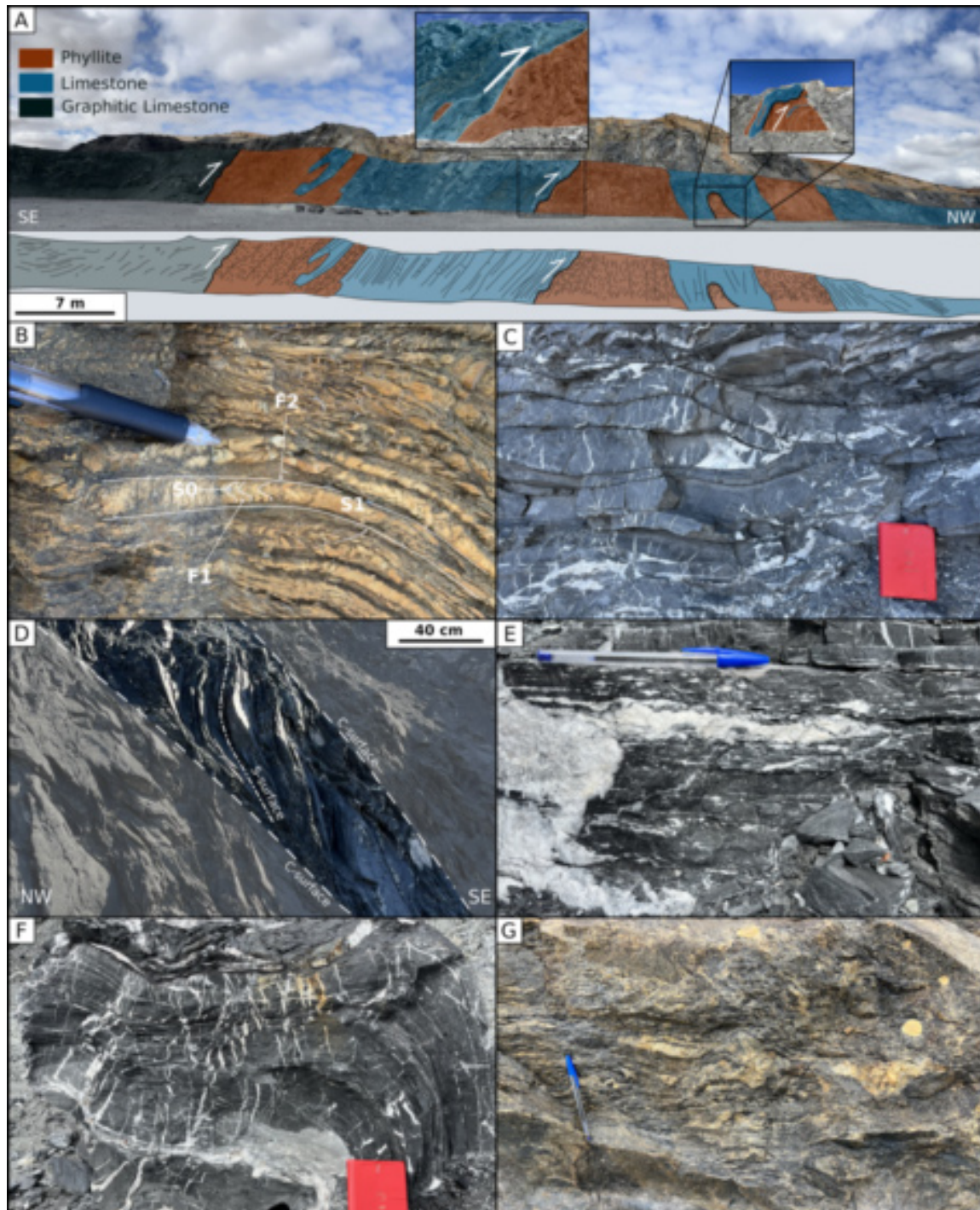


Figure 2. (A) Northwest-southeast oblique section through imbricated phyllite and limestones in the Vondeling quarry, outside Piketberg. Bedding (S_0) is only preserved in some of the massive limestones, but is largely overprinted by the pervasive S_1 cleavage in phyllites and along lithological contacts. (B) Detail of the bedding (S_0) transposition into the S_1 phyllitic cleavage, here in a low-strain domain that preserved the transposition folding (F_1); locality is Bothmaskloof pass, outside Riebeeck Kasteel. (C) Dismembered (boudinaged) bedding in dark-grey limestone units of the lower domain. Quartz-carbonate veins form bedding-normal veins and bedding-parallel veins and quartz-carbonate pods are located in strain shadows and boudin necks, Vondeling quarry. (D) Cross-sectional view of a metre-scale duplex structure ($S-C$) developed in pervasively foliated graphitic limestone, indicating top-to-the-northwest thrust sense of movement, Vondeling quarry. (E) Transposed and boudinaged quartz-carbonate veins define the foliation in a calc-mylonite (below pen). Note the vertical, bedding-normal carbonate veins in undeformed limestone beds above the mylonitic shear zone; longitudinal section (looking northwest, into foliation), Vondeling quarry. (F) Folded and pervasively foliated (S_1) dark limestone with hairline, bedding-parallel carbonate veins are overprinted by steeply-dipping carbonate veins. The later veins can be seen to terminate or being displaced against foliation planes, Vondeling quarry. (G) Strongly-foliated (S_1) biotite gneiss and quartz veins/pods, exposed along a road cut some 8 km northwest of the town of Malmesbury, document the presence of higher-grade metamorphic rocks in the lower phyllite domain, interpreted to be a thrust sliver.

preserving original geometries as well as deformed and tightly folded quartz- and carbonate veins. Undeformed veins in limestones typically form two orthogonal sets. Short, extensional veins normal to bedding terminate against bedding planes and, in many cases, against bedding-parallel veins (Figure 2F). The veins at right angles to bedding indicate a component of vertical, bedding-normal shortening, whereas bedding-parallel veins and the intersection of vein sets point to close-to lithostatic fluid pressures during veining (e.g., Bettelli and Vannucchi, 2003; Raimbourg et al., 2009; Meneghini et al., 2009). In mylonites, high-angle, bedding-normal veins can be seen to become progressively transposed into the mylonitic foliation, corroborating the syntectonic fluid flow.

Distribution of phyllitic rocks

Rocks of the phyllitic domain are confined to the central Swartland region, east of the main Colenso Fault Zone (Figure 1). The phyllites are exposed in the core of the two regional-scale, northwesterly trending and doubly-plunging Swartland and Spitskop domes (F_2 folds, Figure 3). However, isolated outcrops of phyllites indicate a significantly wider distribution of the rocks in the central Swartland region, but the outcrop conditions prevent a precise delineation of the boundary with structurally overlying rocks. Sporadic outcrops of highly sheared quartz-sericite phyllite and intercalated lenses of chlorite schist north of the town of Piketberg indicate the presence of lower-domain rocks north of the Swartland and Spitskop domes (Figure 3). Here, the phyllites are unconformably overlain by the Ordovician Cape Supergroup that forms a prominent outlier (Figure 3). The pervasively foliated and lineated phyllites have previously been mapped as the Piketberg Formation, distinguished by their much higher fabric intensities compared to the underlying, well-bedded, only weakly foliated and openly folded rocks of the Porterville Formation (Hartnady et al., 1974; SACS, 1980; Gresse et al., 2006). The contact between the Piketberg and Porterville formations is not exposed but the sharp contrast in fabric intensities would suggest a structural contact along which rocks of the lower domain were thrust on top of the upper domain (Figure 4). Similarly, large exposures of biotite gneiss (Figure 2G), southwest of the Swartland Dome, document the presence of higher-grade metamorphic rocks in the region (Figure 3). Contacts with the lower-grade phyllites are not exposed, but the pervasive fabric development and tight intrafolial folds indicate the affinity of the gneisses to the phyllitic domain, probably representing a deeper, thrust sliver (e.g., Belcher and Kisters, 2003; Frimmel et al., 2013).

Regional scale strike-slip fault zones

Colenso Fault Zone (CFZ)

The CFZ is the largest fault in the Saldania Belt that can be traced over a length of over 120 km. It is a subvertical, northwesterly-trending and up to 8 km wide, anastomosing fault zone. For the most part, the fault is developed in granites of the CGS and exposures of the fault are best in the central and largely gneissic Darling batholith. In the Malmesbury Group, the trace of the CFZ is only indicated by intermittent

quartz-vein stockworks. The northern extent of the CFZ exposes the damage zone of the CFZ, marked by an up to 3 km wide network of cataclasites in the Vredenburg and Trekoskraal granites (e.g., Schoch, 1972). Along its southern extent, in the Stellenbosch pluton (Figure 1), the fault becomes somewhat indistinct and strain is more distributed as the fault branches into several smaller splays.

The Darling batholith is the only gneissic pluton of the CGS and magmatic and solid-state gneissic fabrics, but also mylonitic and cataclastic shear zones testify to the largely synkinematic emplacement of granites along the CFZ. It is a compositionally heterogeneous, sheeted granite with a distinctly ovoid outcrop (45 x 12 km), parallel to the CFZ (Figure 5). Successively emplaced phases of the Darling batholith show earlier magmatic and high-temperature, northwest-trending solid-state gneissic fabrics over a width of up to 8 km. Fabric intensities are variable and range from the preferred alignment of metasedimentary xenoliths and magmatic enclaves (Figure 6A), feldspar megacrysts (Figure 6B), to banded gneisses (Figure 6C and 6D). The regionally developed fabrics indicate a rather distributed strain that is overprinted by northwest-trending, narrower (5 to 20 m), anastomosing mylonitic shear zones and cataclasites (Figure 6E). In many cases, intrusive granite sheets show markedly higher fabric intensities and mylonitic fabrics compared to the adjacent wall-rock granites and gneisses. This suggests the localisation of strain into the later magmatic phases and underlines the syntectonic emplacement of the granites. Notably, deformation textures and fabrics in the Darling batholith record much higher temperatures of formation compared to the regional, greenschist grades of metamorphism. This also stresses the largely synkinematic emplacement and probably positive feedback between magma emplacement, associated rheological weakening, and strain localisation (e.g., Tommasi et al., 1994; Kisters et al., 2002). U-Pb zircon ages from earlier gneissic granodiorites (547 ± 6 Ma, Da Silva et al., 2000) to late-stage, but mylonitised leucogranites (523 ± 3 Ma) thus document deformation along the CFZ over some 30 to 35 Ma, between ca. 555 and 520 Ma (Kisters et al., 2002).

S-C and S-C' fabric in mylonites and cataclasites indicate mainly sinistral strike-slip kinematics along the CFZ (Figure 6G). Mineral stretching lineations along mylonites, the alignment of K-feldspar megacrysts, and the elongation of enclaves and xenoliths in granites and granite gneisses underline the strike-slip kinematics. Earlier mylonites with sinistral kinematics, are, in places, overprinted by narrow cataclastic bands that indicate a dextral sense of shear (Figure 6H). Similar dextral kinematics have also been described for some of the northern outcrops of the CFZ (e.g., Kisters et al., 2002). The deposition of the coarse-clastic, post Pan-African Klipheuwel Group in half-graben structures (Raath, 2022) indicates the late-stage, post-Pan African reactivation of selected fault strands of the CFZ, particularly along its southern extent, just north of Stellenbosch.

Piketberg-Wellington Fault Zone (PWF)

The Piketberg-Wellington Fault (PWF) is located to the northeast of the CFZ. It is the second-largest fault in the western Saldania Belt and several studies have emphasised its significance for the

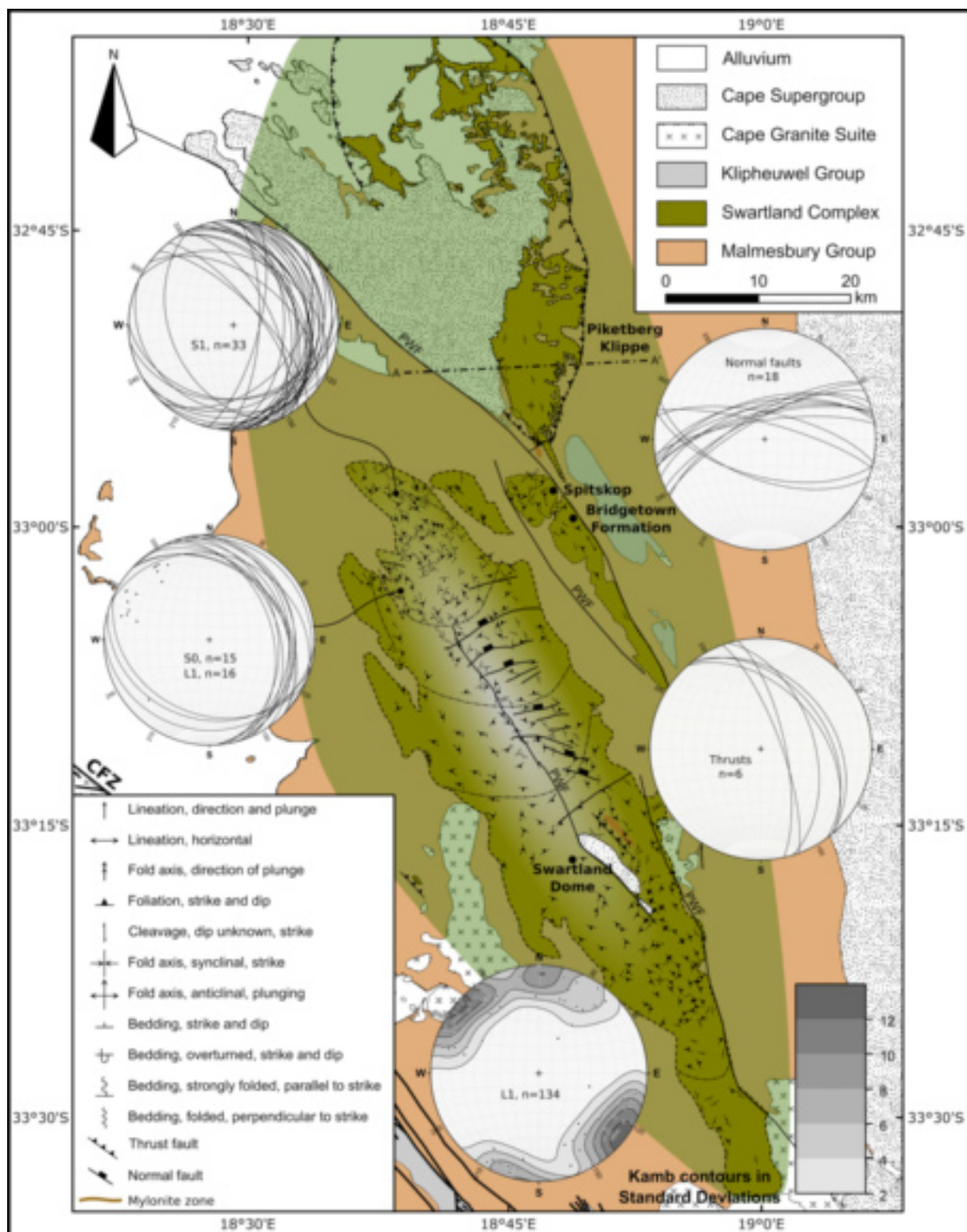


Figure 3. Distribution of lower domain rocks in the Swartland and Spitskop domes. The structural map for the lower domain sees the S_1 foliation defining a rounded trend, following the F_2 Swartland Dome contours, indicated by the structural formlines. The lower domain lineations form a predominantly northwest-southeast trend. Normal faults generally have an east-northeast–west-southwest trend and local thrust planes show northwest-southeast trends. Stereonets show structural trends for the Swartland dome. The green halo surrounding the Swartland Complex indicates the much wider extent of the Swartland Complex than the currently accepted subdivisions. (All stereonet are equal area projections into the lower hemisphere).

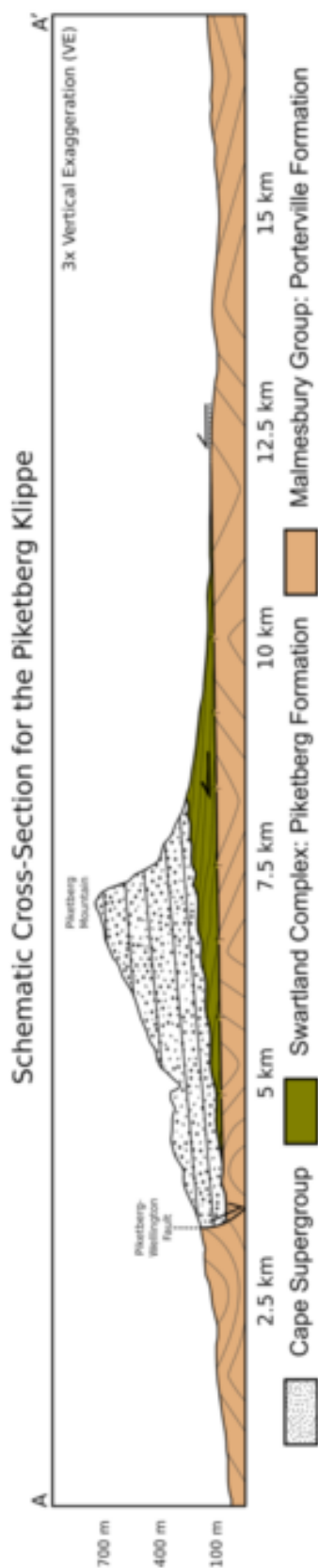


Figure 4. Schematic cross-section through Piketberg Mountain (A-A') showing phyllites of the Piketberg Formation (lower domain) overlying the Porterville Formation (upper domain) — the Piketberg Klippe. The contact between the two units is not exposed but the sharp contrast in fabric intensities indicate a structural contact.

Pan-African evolution of the belt, possibly also as a terrane-bounding fault (Hartnady et al., 1974; Buggisch et al., 2010; Frimmel et al., 2013). However, the actual trace, significance, and timing of the fault are not clear and controversially discussed (e.g., Hartnady et al., 1974; Rabie, 1974; De Villiers, 1972; Slabber, 1995; Belcher and Kisters, 2003; Buggisch et al., 2010; Frimmel et al., 2009; 2013). Along its northern extent, the fault can be traced for some 40 km from Aurora to the town of Piketberg (Figure 1). Here, the fault forms the western faulted contact along which a large outlier of the Paleozoic Cape Supergroup (Piketberg) has been downfaulted. This points to a post-Pan-African, possibly Mesozoic, normal displacement with a downthrow to the northeast (Belcher, 2003). Towards the south, the actual trace of the fault becomes very indistinct. The northwest-trending fault zone that bounds the outlier of Cape Supergroup rocks at Riebeeck Kasteel also shows normal displacement, probably Mesozoic in age. Quartz veining and fault breccias along the trace of the fault are common, but a Pan-African age of faulting cannot be confirmed with certainty. In either case, the surface expression of the PWF is far less prominent and much narrower compared to the CFZ.

Upper domain

Lithological inventory

Rocks of the upper domain encompass a very-low grade metamorphic, but well-bedded, turbiditic greywacke-shale sequence (Figure 7A) (e.g., Hartnady et al., 1974; Von Veh, 1983; Rozendaal et al., 1999; Belcher and Kisters 2003; Gresse et al., 2006; Rowe et al., 2010; Buggisch et al., 2010; Frimmel et al., 2009, 2013; to name but a few). Sedimentary features such as graded bedding, ripple marks, channel fills and rip-up clasts, dewatering and soft-sediment deformation structures are well preserved and testify to the low internal strains of the rocks (Von Veh, 1983; Theron et al., 1992; Gresse et al., 2006; Frimmel et al., 2013). Minor volcanic rocks (felsic tuff, amygdaloidal lava, volcanoclastic rocks) are intercalated with the metaturbidites in the Bloubergstrand Member (Von Veh, 1983; Kisters et al., 2015), and pervasively chloritised intrusive sills are locally developed. Zircons from a felsic tuff unit at Bloubergstrand yields U-Pb crystallisation ages of ca. 555 Ma, constraining the timing of sedimentation to only shortly before the emplacement of the earliest phases of the CGS at ca. 552 Ma (Kisters et al., 2015). The crystallisation age correlates with the youngest detrital zircon ages of ca. 560 Ma for the metasediments (Armstrong et al., 1998; Frimmel et al., 2013).

Structural geology

Rocks of the upper domain rocks are only deformed by one fold generation, here designated F_2 . F_2 folds are non-cylindrical folds with overall northerly to northwesterly trends and gentle plunges ($<15^\circ$) to the northwest and/or southeast (Figure 7B). Fold axial traces and bedding formlines show northwest trends in the southern and central parts of the belt, closer to the central CFZ, but are more northerly in the north and far south (Figure 8). Fold axial traces in the south show a distinct

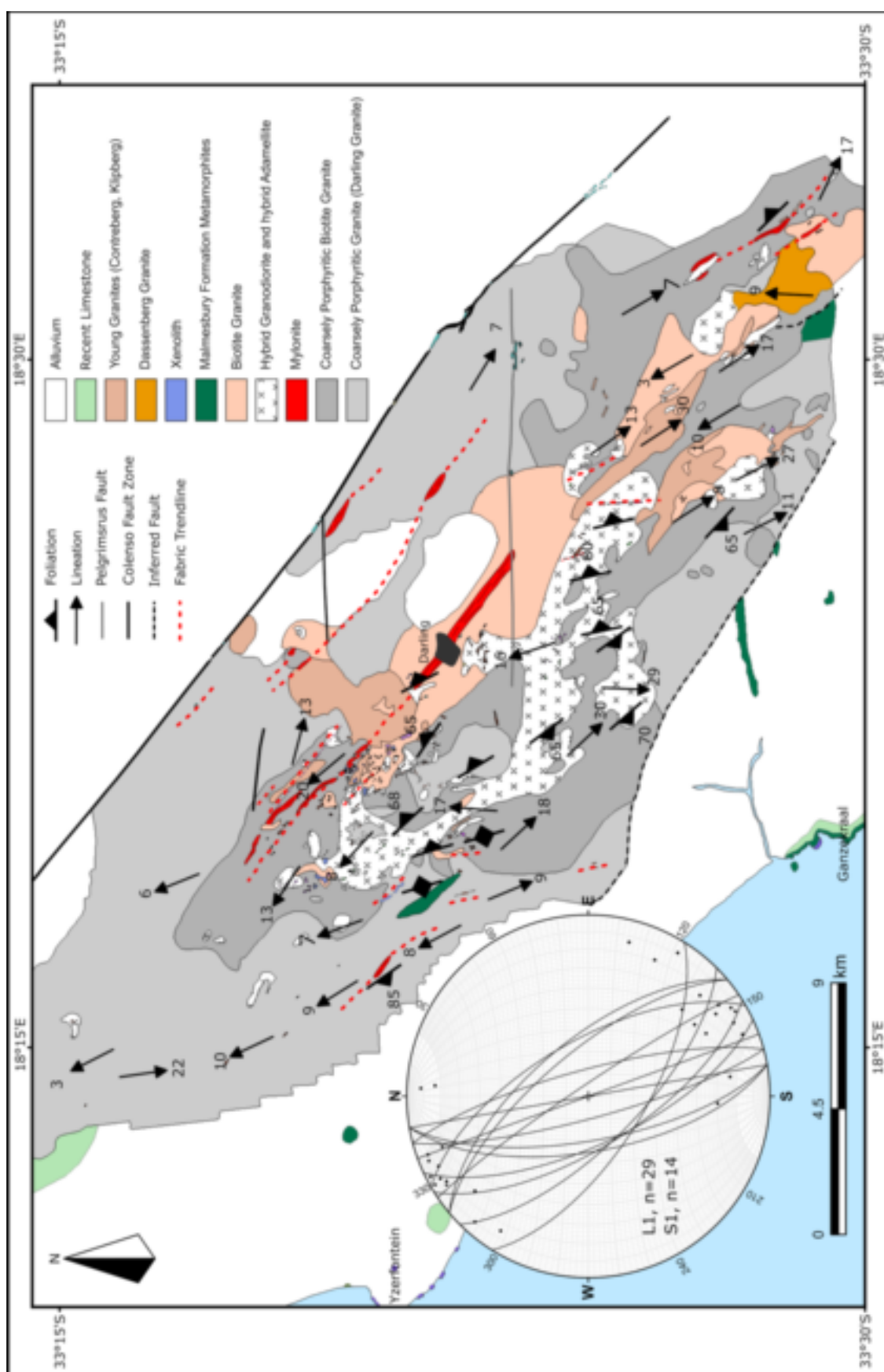


Figure 5. Simplified geological map of the Darling batholith showing the distribution of the main granite phases (after Schoch, 1972). Magmatic and solid-state fabrics (planar and linear, stereographic projection, inset) are developed throughout the batholith and in different intrusive phases. High-strain zones (mylonites and cataclastics) are more localised, showing northwest strikes, parallel to the Colenso Fault Zone (CFZ). The northeastern parts of the batholith are thought to be bound by the core of the CFZ, but are only poorly exposed.

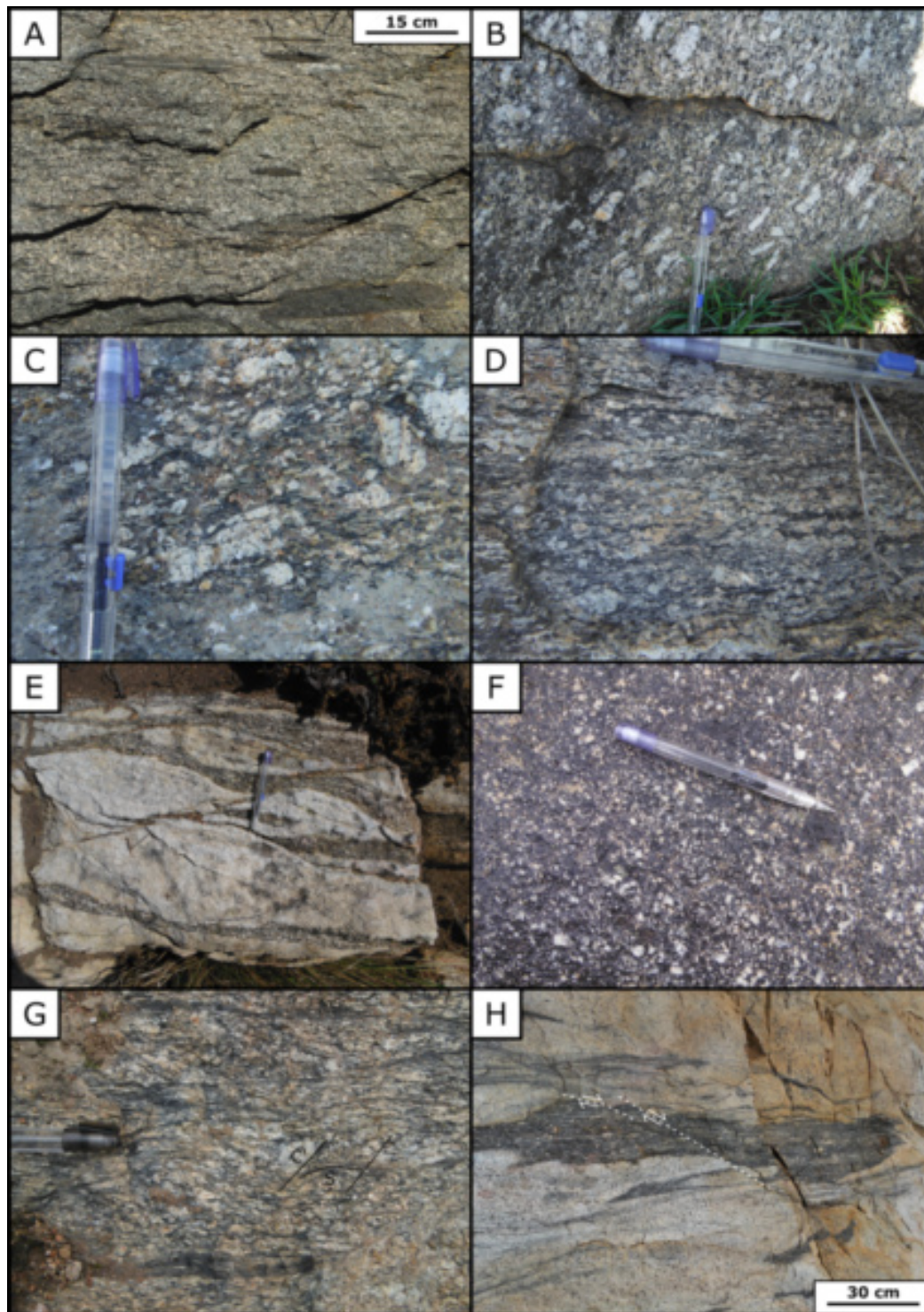


Figure 6. (A) Close-up (plan-view) of the granodioritic phase of the Darling batholith containing abundant and highly stretched metasedimentary xenoliths defining the steep northwest-trending planar fabric. (B) Cross-sectional view of the steep magmatic fabric in a porphyritic phase of the Darling granite defined by the preferred orientation of K-feldspar megacrysts. (C) Close-up (plan view) of protomylonitic fabric development in porphyritic granite, showing quartz ribbons wrapping around marginally recrystallised K-feldspar megacrysts. (D) Mylonitic fabric overprinting originally megacrystic Darling granite (plan view). (E) Lit-par-lit intrusive relationships between late-stage leucogranite and earlier granodiorite, both mylonitised and overprinted by sinistral shear bands (S-C') (plan view). (F) Cataclastically deformed granodiorite of the Darling batholith. Note the fragmented relics of original feldspar (white) (plan view). (G) Close-up (plan-view) of sinistral shear-sense indicators (S-C fabrics) in mylonitic Darling granite overprinted by later ultracataclasites (black, foliation-parallel streaks) illustrating the transition from high-temperature ductile to lower-temperature brittle fabrics during cooling of the granite. (H) Oblique view of a pervasively mylonitised, earlier intrusive phase of granodiorite, forming a xenolith in mylonitised leucogranite. Mylonitised granites show sinistral strike-slip kinematics, but are deformed by later cataclastic shear bands showing dextral kinematics (annotated).

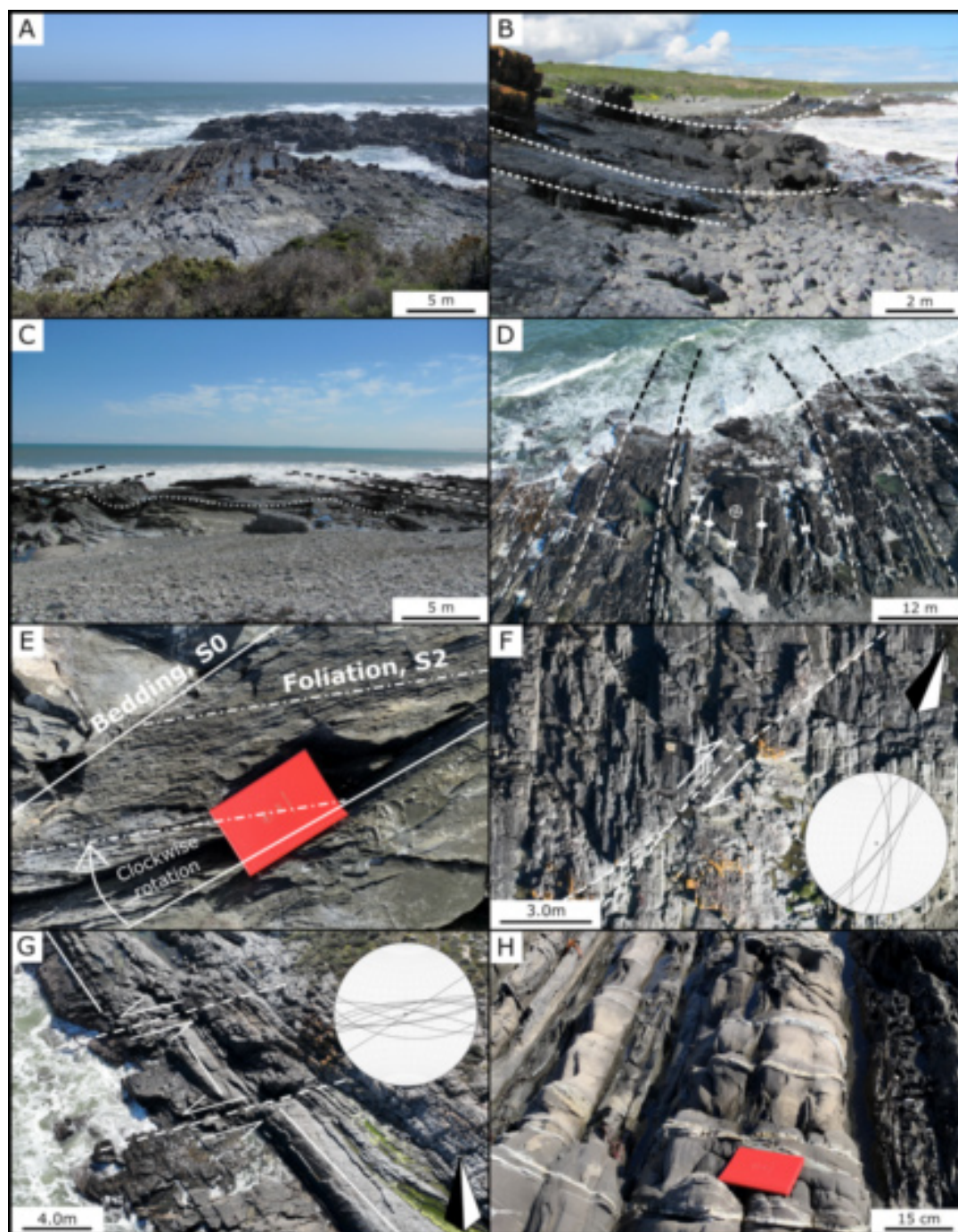


Figure 7. (A) Coastal outcrop of well-bedded, steeply-dipping metaturbidites of the upper domain (Tygerberg Formation along the Atlantic coastline near Ganzekraal, looking northwest). (B) Open folding (F_2 , S_0 annotated) of well-bedded black shale units, illustrating the deformation of folded regions of upper-domain rocks, Grotto Bay, looking southeast. (C) Coastal outcrop, looking northwest, illustrating the presence of steep, isoclinally folded domains (left- and right-hand side of photo, S_0 annotated) enveloping gently folded bedding in black shales (stippled). This pattern is repeated on a kilometre-scale and seems characteristic of F_2 folding in the upper domain. (D) Bird's-eye view of the outcrop shown in Figure 7C, bedding and dip and strike of S_0 annotated by dashed lines, fold hinges indicate the gentle folding of the subhorizontal strata. (E) Plan view of the S_2 transecting cleavage in black shale of the upper domain (coastal outcrop of the Tygerberg Formation). The S_2 cleavage shows a consistently clockwise rotation with respect to F_2 fold axial planes throughout F_2 folds. (F) Plan view (drone image) of northeasterly-trending (stereonet with great circles) narrow, dextral fault in the upper domain, Ganzekraal locality. (G) Oblique view (drone image) of easterly-trending (stereonet with great circles) sinistral faults. The sinistral and dextral faults (Figure 7F) define a conjugate pattern, indicating their formation during northeast-southwest shortening and northwest-southeast stretch; coastal outcrops at the Ganzekraal locality. (H) Oblique plan-view of interbedded greywacke and shale containing a closely-spaced, subvertical, northeasterly-trending, extensional (mode I) quartz vein set in greywacke emplaced normal to the northwest-trending fold hinge, indicating a fold-hinge parallel (northwest-southeast) stretch during folding, Ganzekraal locality.

anticlockwise rotation from northerly towards more northwest trends closer, and in parallelism, to the CFZ and consistent with the sinistral sense of shear along the fault (Figure 8). Notably, a similar rotation of fold traces is not recorded along the PWF in the north. The curvilinear formlines in parts of the central Swartland region reflect the refolding of the S_0/S_1 transposition fabrics of lower domain rocks by F_2 folds (Swartland and Spitskop domes) (Figure 3).

The majority of folds are steeply inclined (60 to 80°), southwest verging, or more or less upright folds. There is no systematic change of fold vergence directions across the belt and vergence direction may change even in individual folds along strike, underlining the non-cylindrical geometry of F_2 folds. First-order folds have widths of 1.5 to 2 km, and those of 2nd- and 3rd-order folds range from 70 to 300 m and 30 to 70 m, respectively. While first-order F_2 folds show consistently gentle-to open interlimb angles (85° to 130°), fold interlimb angles of second- and third-order folds range from isoclinal to open (0° to 110°). These variations in fold tightness are characteristic of F_2 folds, both across strike and in adjoining fold pairs and/or along strike and parallel to the axial trace of individual folds. On a more local scale (10's to 100's of metres), gently folded strata can be seen to abruptly abut against domains of vertically dipping, tight-to isoclinally folded rocks (Figure 7C). These zones of upright, isoclinal folding describe an anastomosing pattern enveloping gently folded domains, resembling anastomosing shear zones wrapping around lithons (Figure 7D). However, there is no evidence of non-coaxial strains in rocks contained in the vertical domains and rocks have experienced only very little internal deformation despite the isoclinal folding. This pattern of steep domains enveloping openly folded regions is repeated on a regional scale. Kilometre-wide domains of open folding (e.g., a first-order synform underlying the area at Melkbosstrand, Figure 1) alternate with upright, tightly folded, and strongly foliated rocks (e.g., at the Bloubergstrand locality, Figure 1). The only intermittent and limited exposures along the coast do not allow for a more detailed delineation of geometries.

F_2 folds are associated with a steep- to subvertical S_2 foliation. The foliation is well developed in tightly-folded, shaley units, while greywacke beds show little evidence of the S_2 foliation other than a spaced pressure-solution cleavage. The S_2 foliation is a transecting cleavage, as pointed out by Rowe et al. (2010), with a consistent clockwise rotation of the foliation with respect to the fold-axial plane by between 5 to 15° (Figure 7E) (Kisters and Belcher, 2018). The transecting cleavage and consistent clockwise deflection can be interpreted to indicate F_2 folding during sinistral transpression (Soper et al., 1987; Fossen, 2010).

Linear fabrics are rare in upper-domain rocks and mainly comprise northerly- to northwesterly-trending and gently plunging intersection lineations (S_0/S_2). Slickenside lineations on bedding planes normal to or at high angles to fold hinges are mainly developed in greywacke units and relate to flexural slip during F_2 folding.

The coastal exposures show numerous small-scale, mainly strike-slip faults cross-cutting rocks of the upper domain at high angles. The faults are commonly steep (>70 to 80°), and narrow (<1 to 2 m-wide) with strike extents of tens- to hundreds of

metres. The presence of larger, kilometre-scale faults is indicated by their morphological expression and displacement of lithological units. Most faults are brittle-ductile faults, containing brecciated or sheared quartz veins in the fault core and a greenish chlorite alteration, particularly in shaley units. The alteration and ductile component of deformation indicate that the structures are Pan-African in age, having formed during low-grade metamorphic conditions and not related to younger events. Moreover, the faults show a systematic arrangement around F_2 folds that suggests a relationship between the regional F_2 folding and faulting. Broadly easterly trending, steep northerly dipping sinistral faults form parallel, closely-spaced arrays at high-angles to F_2 fold hinges. Displacements along individual fault planes range from 2 to 5 m. North to northeasterly trending faults with dextral displacement (Figure 7F) and east-southeasterly faults with sinistral displacement (Figure 7G) enclose an acute angle of some 60 to 70° and form a conjugate set of faults around F_2 folds. The conjugate fault sets are consistent with fold-hinge parallel stretch, indicating northwest-southeast extension and northeast-southwest compression. Fold hinge/bedding parallel faults are common in the southernmost exposures of the belt in the area around Strand (Figure 1).

Hydrothermal veining

Quartz veins in rocks of the upper domain are largely confined to more competent greywacke units. Unlike the transposed veins in lower-domain rocks, quartz-vein geometries are well preserved, and two main quartz-vein sets show remarkably consistent orientations that point to the regional controls of quartz-vein formation. Extensional (mode I) quartz veins are typically syntaxial veins made up of fibrous quartz intergrown with dark-green chlorite. Vein thicknesses range from hairline fracture fillings to >25 to 30 cm. Veins of this vein set are subvertical (dips >80°) with northeasterly strikes, normal to bedding and F_2 fold hinges (Figure 7H). For the most part, veins are undeformed and indicate late-stage veining during the advanced stages of fold tightening. A second vein system comprises a conjugate set of steeply-dipping shear veins made up of a more prominent east-northeasterly trending vein set, with a sinistral sense of displacement, and a less pronounced north-northeasterly trending dextral set. Overprinting relationships with extensional (mode I) quartz veins and chlorite alteration along the conjugate veins suggest the more or less contemporaneous evolution of both vein systems. In this orientation, both vein systems have accommodated a component of fold-hinge parallel (northwest-southeast) stretch during regional F_2 folding and shortening. Notably, quartz-vein orientations in the greywacke units are consistent with regard to F_2 fold-hinge orientations throughout the region and variations in vein orientations relate to changes in fold-hinge orientations.

Distribution

Rocks discussed here as part of the upper domain correspond to the Tygerberg Formation of currently accepted stratigraphic subdivisions (Hartnady et al., 1974; SACS, 1980; Gresse et al.,

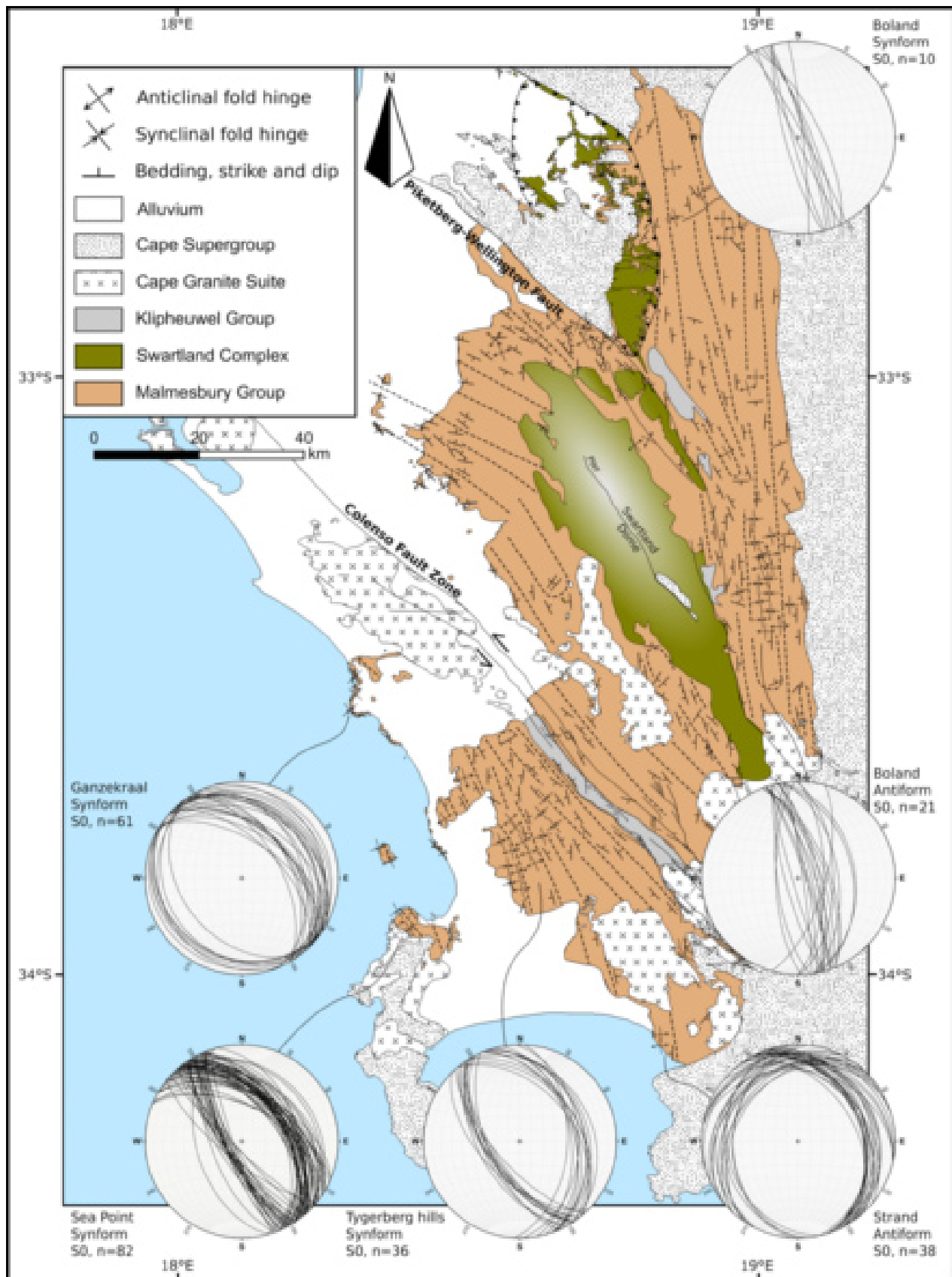


Figure 8. Structural formline map of bedding (dashed lines) and F_2 fold hinges in upper-domain rocks (Malmesbury Group). Note the rotation of formlines from northerly trends in the north (Boland) and far south (Strand) to more northwesterly trends, parallel to the Colenso Fault Zone (CFZ) in the central parts.

2006), southwest of the CFZ, but also large parts of the Moorreesburg and Porterville formations to the northeast in the Swartland and Boland regions (Figure 8). Kisters and Belcher (2018) referred to these rocks as the Malmesbury Group and likened them to deposits of a fore-arc basin. The term Malmesbury Group underlines the regional extent of the rocks and across the supposedly terrane-bounding Colenso and Piketberg-Wellington faults but contrasts them with phyllitic rocks of the mélange-like Swartland Complex, i.e., lower domain rocks. Clear distinctions between upper and lower domain rocks in the field are the lack of the pervasive low-angle (S_1) phyllitic cleavage and excellent preservation of primary bedding features (Figure 7A), the scarcity of linear fabrics, and the lower abundance of quartz veins in the former. The contact between upper and lower domain rocks is nowhere exposed. Notably, detrital zircon ages for the upper domain and lower phyllitic domain are indistinguishable (Frimmel et al., 2013).

Discussion

The regional mapping and integration of structural results for the western Saldania Belt delineate distinct structural domains in the belt. Importantly, different fabric domains formed broadly contemporaneous and record Pan-African strains. Deformation occurred over a period of some 50 Ma, between at least *ca.* 555 Ma, the timing of deposition of rocks of the upper and lower domains (e.g., Armstrong et al., 1998; Frimmel et al., 2013; Kisters et al., 2015) and *ca.* 510 Ma, the emplacement of the post-tectonic, A-type Klipberg Granite (e.g., Chemale et al., 2011; Scheepers and Schoch, 2006) that sharply cross-cuts regional folds and fabrics. Later tectonic overprints of the Cape Fold Belt or Mesozoic rifting are minor and can clearly be distinguished from the earlier Pan-African strains.

Southeast-directed oblique convergence and sinistral transpression during the Pan-African amalgamation of southwest Gondwana have been described for the Kaoko and Gariep belts (Gresse, 1994; Goscombe et al., 2003; Frimmel and Fölling, 2004; Konopásek et al., 2005) and have also been postulated for the Saldania Belt, although poor exposure has hitherto been the main limiting factor in the formulation of more widely accepted, regional tectonic models (e.g., Hartnady et al., 1974; Rozendaal et al., 1999; Rowe et al., 2010; Frimmel et al., 2011, 2013; Kisters and Belcher, 2018). The kinematics of transpressional deformation are, amongst other factors, determined by the obliquity of the plate margin with respect to plate movement, or the angle of convergence (e.g., Fossen and Tikoff, 1993; Teyssier et al., 1995; Jiang et al., 2001). Importantly, during oblique convergence and collision, the vector of plate motion is rarely parallel to the shortening direction recorded in rocks of the overriding plate. In cases where the angle between the plate margin and the vector of plate movement is $>20^\circ$, deformation is pure-shear dominated and horizontal shortening is associated with the vertical extrusion of rocks and finite strains result in vertical stretching lineations. At convergence angles $<20^\circ$, theoretical models indicate a switch from pure shear to strike-slip dominated transpression. At these lower angles of

convergence, horizontal shortening is associated with a horizontal stretch parallel to the continental margin and horizontal stretching lineations (Fossen and Tikoff, 1993). Many obliquely convergent margins record more complex three-dimensional strains. These strain patterns document the partitioning of transpressional strains into discrete, margin-parallel shear zones that accommodate the margin-parallel component of the oblique convergence and broader, folded domains that accommodate the margin-normal shortening component (e.g., Tikoff and Teyssier, 1994; Barnes et al., 1998; Wallace et al., 2004; Conand et al., 2020; Greve et al., 2020). Strain partitioning commonly results from the localisation of strain into crustal heterogeneities, such as pre-existing structures, rheologically weaker lithologies, or thermal weakening effects around plutonic complexes or volcanic arcs (e.g., Jones and Tanner, 1995; Saint Blanquat et al., 1998).

We interpret the different fabric domains, strains, and kinematics in the western Saldania Belt as the result of a pronounced strain partitioning during the Pan-African oblique convergence and closure of the Adamastor Ocean, similar to that of the northern belts. Despite this, particular boundary conditions specific to the Saldania Belt and outlined below, account for the different tectonometamorphic evolution and setting of the belt. Previous studies (e.g., Belcher, 2003; Rowe et al., 2010; Kisters and Belcher, 2018) have likened the juxtaposition of these domains to a section through a fore-arc region, from the deeper accretionary prism (lower domain, or Swartland Complex) into the overlying fore-arc basin (upper domain, or Malmesbury Group). This was based on the structural style and lithological assemblages and tentative correlation of units, but without a regional analysis and integration of kinematics and strains. This also implies that kinematics and fabric domains in the western Saldania Belt record the partitioning of strain between two structurally overlying levels during oblique convergence.

Deformation of upper-domain rocks

F_2 folds throughout the western Saldania Belt show northerly to northwesterly trends, parallel or at low angles to the continental margin (Figure 9A). The transecting S_2 foliation is commonly steep and indicates folding during subhorizontal shortening, but with a component of sinistral transpression, although distributed and not localised. Even the belt's tight- to isoclinal folds that envelope the more openly folded F_2 domains do not show evidence of non-coaxial strains. Quartz-vein sets in rocks of the folded Malmesbury Group are ubiquitous. For the most part, the veins are undeformed and have formed late during folding, probably during the tightening of folds and fold-lock-up stage. The veins record infinitesimal strains that can be correlated with the actual stresses during folding and vein formation (e.g., Teyssier et al., 1995). Extensional veins in the folded strata form prominent sets normal to the hinges of F_2 folds and shear veins form a conjugate set around extensional veins. In this orientation, F_2 folds and associated quartz vein sets indicate folding during east-northeast–west-southwest shortening, nearly normal to the continental margin, and a fold-

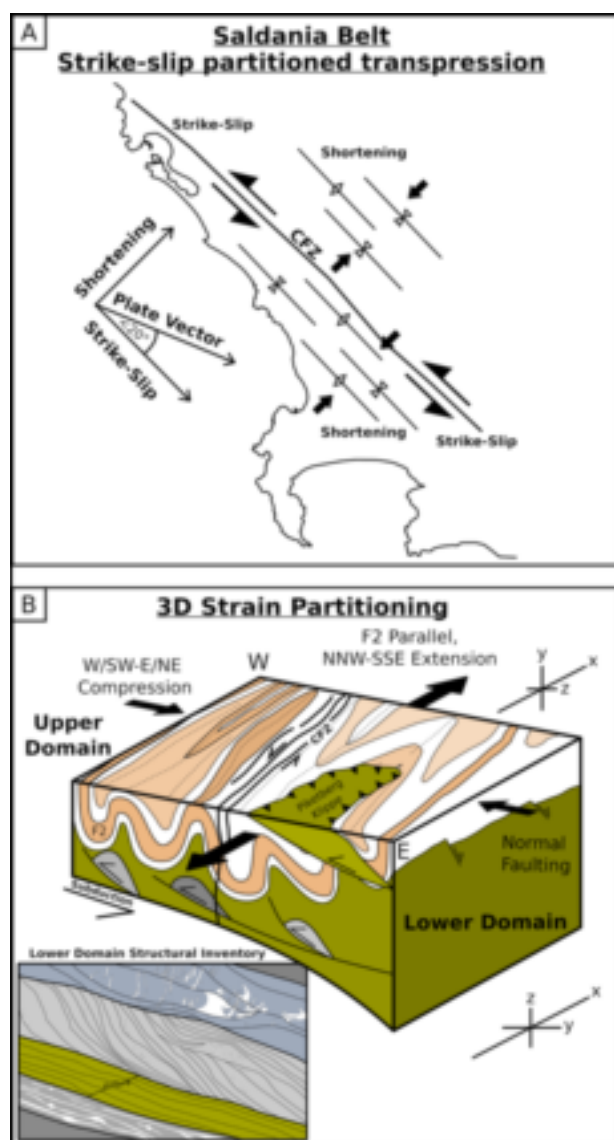


Figure 9. (A) Schematic sketch illustrating the strike-slip partitioned transpression discussed here for the Saldania Belt. Low angle ($<20^\circ$) convergence is partitioned into (1) a margin-parallel sinistral strike-slip component along the Colenso Fault, and (2) a margin-normal, coaxial-shortening component in intervening domains dominated by F_2 folds. (B) Schematic 3D block model through the upper and lower domain in the central parts of the Saldania belt, illustrating the main structural components discussed in the paper. X, Y, and Z ($X \geq Y \geq Z$) indicate the finite strain axes recorded for upper and lower domain rocks. Note the orogen-parallel stretch (X) in both domains, but the swap of the shortening (Z) and intermediate strain (Y) axes between the upper and lower domain. Inset shows a schematic summary of the main fabric elements in the lower domains.

hinge parallel, north-northwest–south-southeast stretch, parallel to the margin of the Saldania belt (Figure 9). The orientation and kinematics of conjugate brittle faults at high angle to F_2 folds are consistent with the high-angle shortening and fold-hinge parallel stretch (Figure 9).

The CFZ, in contrast, records mainly sinistral strike-slip kinematics and gneisses, mylonites, and cataclasites show shallowly plunging, northwest-southeast trending stretching lineations. Importantly, the CFZ is, for the most part, developed in synkinematic granites of the CGS, from the central Darling batholith and the Stellenbosch pluton in the south to the Vredenburg adamellite and Trekoskraal granite in the north. This spatial and temporal relationship underlines the positive feedback loop between granite emplacement, heat advection, and strain localisation that has promoted strike-slip partitioning into the CFZ. Strain localisation into granite phases is clearly evident on an outcrop scale in the Darling batholith, where even the youngest (*ca.* 520 Ma) granite sheets are mylonitised (Figures 5 and 6). High-temperature deformation fabrics in the granitic gneisses compared to the regional lower-greenschist facies metamorphism also indicate the advection of heat associated with the emplacement of the magmas and probably into the surrounding rocks. This is likely to have contributed further to the weakening of the shallow crustal section. The rotation of F_2 folds into parallelism with the CFZ on either side of the fault (Figure 9A) is consistent with the sinistral strike-slip kinematics. The rotation may indicate that strike-slip shearing outlasted F_2 folding or possible higher strain rates along the CFZ.

In contrast to the CFZ, the PWF is not associated with any granites of the CGS, its expression is far less prominent and strike-slip kinematics are not evident along the fault zone. Instead, the down-faulting of the Cape Supergroup illustrates normal kinematics. On a regional scale, F_2 fold hinges seem unaffected by the PWF, unlike the CFZ. Taken in conjunction, the PWF is unlikely to have represented a major Pan-African structure and it is a younger structure of probably Mesozoic age.

In summary, the broad regions of distributed folding (F_2) and the northwesterly trending, sinistral strike-slip CFZ document the pronounced partitioning of strain into a margin-normal, largely coaxial component (Z of the finite strain ellipsoid is horizontal and strikes east-northeast–west-southwest, with $X \geq Y \geq Z$) and margin-parallel, non-coaxial component of deformation and horizontal stretch of rocks (X of the finite strain ellipse is horizontal and strikes north-northeast–south-southwest) of the upper domain (Malmesbury Group) during oblique convergence (Figure 9). Sinistral transpression was strike-slip dominated, which points to a low angle ($<20^\circ$) of broadly northwest-southeast convergence. Despite this, the consistently developed transecting nature of the S_2 cleavage in folded domains documents that the strike-slip component was not completely partitioned into the CFZ, but more distributed and also developed in the folded domain.

Deformation of lower domain rocks

While detrital zircon ages from rocks of the upper and lower domains document deposition of the rocks at roughly the same time (Frimmel et al., 2013), fabric development, strains, and the orientation of structures are distinctly different (e.g., Belcher, 2003; Kisters and Belcher, 2018). In addition, lower domain rocks contain a larger diversity of rock types, from the omnipresent quartz-sericite phyllites, of likely turbiditic origin,

to metamafic rocks (lenses of chlorite schist and the Bridgetown Formation), limestones, graphitic limestones, as well as chert and higher-grade gneisses. Fabric (S_1) development and bedding transposition document high finite strains in phyllites that have largely obliterated the strain path of the rocks. The more competent limestones, in contrast, preserve structural details that track the incremental evolution and progressive deformation of the rocks. Vertical, bedding-normal extensional (mode I) veins and bedding-parallel boudinage document the vertical shortening of the succession. The vertical vein sets are progressively overprinted and transposed into bedding-parallel mylonites that record top-to-the west and -northwest thrust sense of movement (see also Belcher, 2003; Kisters and Belcher, 2018). Fabric development, thus, records episodes of vertical, coaxial shortening (Z of the strain ellipsoid is vertical) and associated veining, probably as a result of gravitational loading due to thrusting and thickening, alternating with episodes of non-coaxial, bedding-parallel thrusting. The abundance of hydrothermal veins throughout rocks of the lower domain emphasises the role of fluids and deformation in the presence of close-to-lithostatic fluid pressures, as it is widely recorded for accretionary prisms undergoing dewatering and dehydration reactions during loading and prograde metamorphism (e.g., Moore and Vrolijk, 1992; Meneghini et al., 2009). Voluminous syntectonic fluid flow and associated alteration are also likely to have resulted in the pervasive sericitisation of the rocks and formation of the thick and seemingly homogeneous phyllite packages that dominate the lower domain.

Distribution and contact relationships

The previous discussion also highlights those structures and fabrics assigned to distinct phases of deformation (D_1 , S_1 , F_1 , L_1 of the lower domain and D_2 , S_2 , F_2 , L_2 , etc. of the upper domain) have formed at broadly the same time. Their D_1 and D_2 notation

merely reflects the absence of low-angle D_1 fabrics in upper-domain rocks and their formation at different structural levels that record different strains in different parts of the deforming orogenic wedge. This also implies the decoupling of the two domains. The contact between the upper and lower domain is not exposed and the nature of this contact has variably been interpreted as an unconformity or a structural detachment (e.g., Kisters and Belcher 2018). Previous regional studies have mapped lower-domain rocks mainly in the core of the F_2 Swartland and Spitskop domes that expose deeper structural levels (e.g., Newton, 1966; Rabie, 1974; Hartnady et al., 1974; Theron et al., 1992; Blecher 2003). Isolated outcrops mapped in this study suggest a much wider distribution in the central Swartland region (Figure 3) outside the first-order F_2 antiforms, including the Bridgetown Formation. It is conceivable that tectonic underplating at the base of the accretionary wedge or return flow of rocks towards the continental backstop may have led to the regional uplift and exhumation of the structurally deeper levels in the central parts of the prism, in the Swartland region (Figure 10). Thrusting in this part of the lower domain is indicated by the presence of higher-grade gneisses (Figure 6G). Notably, northeast-trending, late-stage normal faults are largely restricted to the lower domain (Figure 3). The orientation and kinematics of the normal faults are consistent with the overthickening and gravitational collapse of the uplifted rocks (Figure 10).

The position of the highly-strained (S_1 , L_1), phyllitic and lithologically heterogeneous Piketberg Formation (Figure 4) overlying folded (F_2) rocks of the Porterville Formation suggests that the former presents a klippen structure bounded by a low-angle thrust (Figure 3). This implies the transfer (thrusting) of lower-domain rocks into the overlying, folded fore-arc basin stratigraphy as it is similarly documented from accretionary prisms of recent convergent margins (e.g., Schlüter et al., 2002; Berglar et al., 2017).

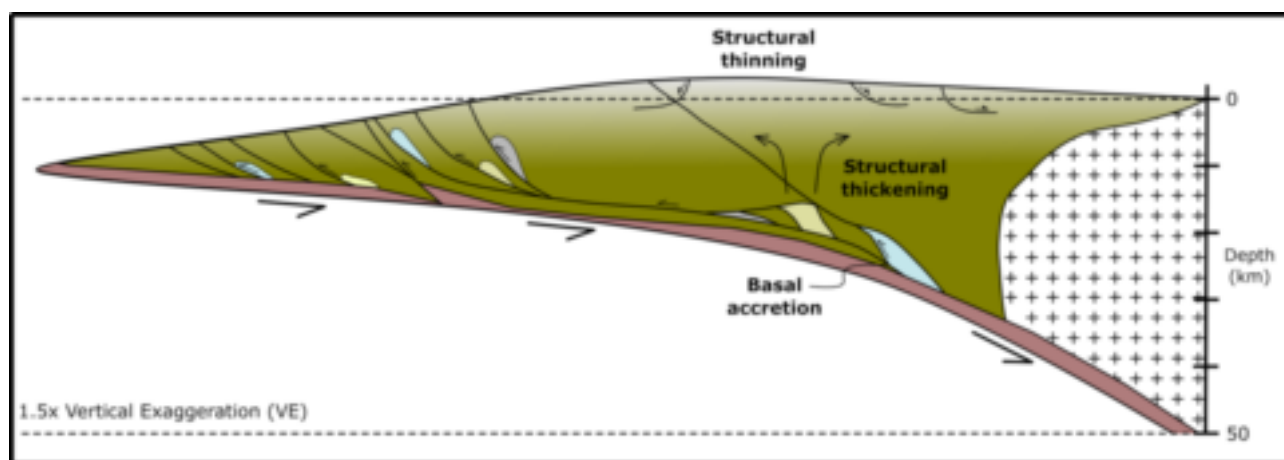


Figure 10. Tectonic underplating at the base of the accretionary wedge or return flow of rocks towards the continental backstop is suggested to have caused the regional uplift and exhumation of the structurally deeper levels in the central Swartland region. East-northeast-trending, late-stage normal faults are interpreted to indicate the overthickening and collapse of the uplifted rocks. Note the east-northeast-trending normal faults are sub-perpendicular to the stretching lineations.

Regional implications

Most regional models regard the Saldania Belt as the southern continuation of the Gariep back-arc basin invoking a direct correlation of structural and lithological units between the two belts (e.g., Rozendaal et al., 1999; Rapela et al., 2011; Frimmel et al., 2009; 2013). However, strains and kinematics reported here correlate with a fore-arc setting of the belt. Moreover, the likely thickness (>20 km) of the Pan-African sediments and the close temporal succession of sedimentation, deformation, metamorphism, and granite plutonism are uncharacteristic for a back-arc setting (see summary in Kisters and Belcher, 2018). If a fore-arc setting for the Saldania belt is assumed, southeast-directed subduction underneath the Cuchillo-Dionisio-Pelotas arc forming the Marmora back-arc basin of the Gariep Belt in the north (Basei et al., 2005; Frimmel et al., 2009, 2013) would cut back eastward and beneath the continental margin of the Kalahari Craton (Figure 11).

The late-stage reversal of strike-slip kinematics, from earlier sinistral to later dextral movement along the CFZ (e.g., Schoch, 1972; Kisters et al., 2002) is particularly prominent along the northern outcrops of the CFZ, but also evident in the Darling batholith (Figure 6H) and implies a change of regional plate

kinematics. In a fore-arc setting, this reversal may be explained by the subduction of the mid-ocean ridge during oblique convergence (Figure 11A and 11B). Initial sinistral transpression records the oblique, southeast-directed subduction of oceanic crust (Figure 11A). Ridge subduction during closure of the Adamastor Ocean below the leading edge of the Kalahari Craton (Figure 11B) then leads to the reversal from sinistral to dextral strike-slip kinematics. This aspect must remain speculative, but it provides a viable explanation for the reversal of strike-slip kinematics in the scenario of oblique subduction. Notably, the very low grades of metamorphism and only limited exhumation of the Pan-African rocks suggest that the Saldania Belt has only experienced a soft collisional event, without the crustal thickening and subsequent isostatic uplift associated with a hard collision.

Conclusion

Regional mapping and the integration of structural data demonstrate that different domains and structural levels in the Saldania Belt can be integrated into a kinematic model of strike-slip partitioned transpression that records the oblique closure of the Adamastor Ocean during subduction below the leading edge of the Kalahari Craton between *ca.* 555 and 520 Ma.

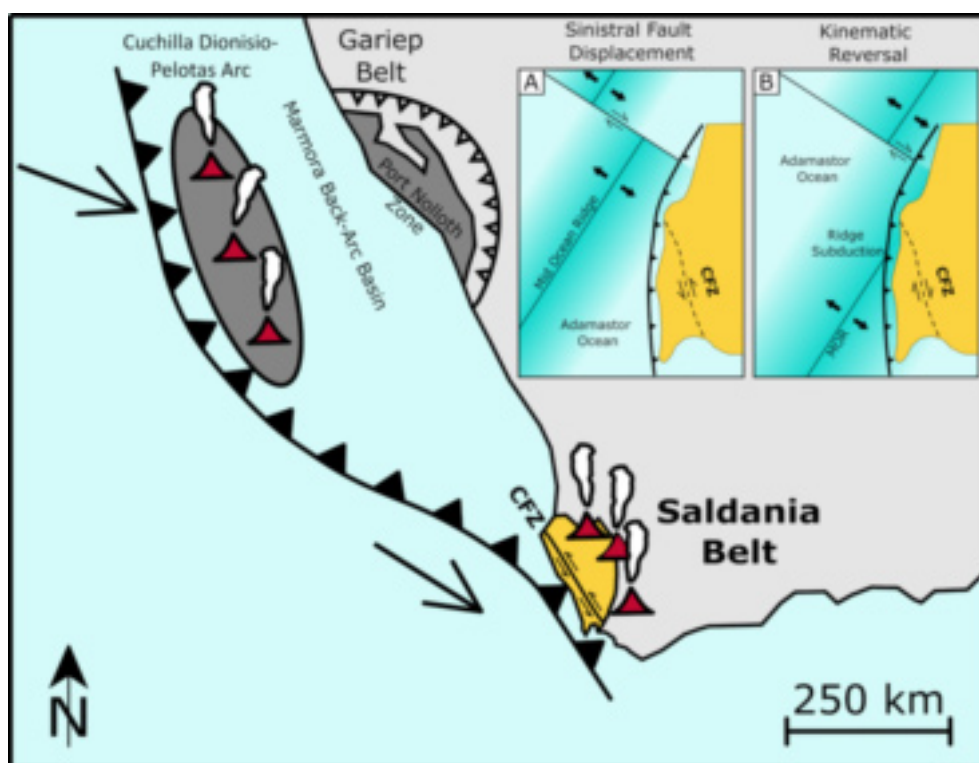


Figure 11. Schematic sketch illustrating the trace for the Pan-African subduction from the Gariep Belt, where subduction is below the Cuchilla-Dionisio-Pelotas arc, suggesting the deposition of the Gariep Supergroup in a back-arc basin (e.g., Basei et al., 2005). Towards the south, in the Saldania Belt, subduction is suggested to have occurred below the continental margin of the Kalahari Craton. The magmatic arc of the subduction is envisaged to be located mainly to the east indicated by the Robertson Pluton and Brewelskloof andesites, but now largely covered by the Cape Supergroup. Insets: (A) The main phase of sinistral transpression records the oblique, southeast-directed subduction of the Adamastor Ocean below the Kalahari Craton; (B) Ridge subduction during oceanic closure leads to a reversal from sinistral to dextral strike-slip kinematics recorded as later-stage overprints along the Colenso Fault.

Transposed and mélange-like rocks in the lower parts of the belt represent imbricated lithologies of an accretionary prism (lower domain). Well-bedded and folded metaturbidites are structurally overlying the prism forming the folded fore-arc basin fill (upper domain), representative of a section through a fore-arc region. This model of a deformed fore-arc region along the leading edge of the Kalahari Craton has important implications for regional reconstructions. The model agrees with the sinistral transpressional deformation recorded in Pan-African coastal belts to the north during the later stages of Gondwana amalgamation, but questions the widely held view of stratigraphic and structural correlations particularly with the neighbouring Gariep Belt.

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